

Progress Report

Title:	Climate-Smart Cotton: Developing Precision Regenerative Practices and Market Opportunities for Addressing Climate Change in the Cotton Belt		
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Funding Source	Non Formula	Reporting Frequency	Annual
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Program Code: A9201**Program Name:** Sustainable Agricultural Systems**Project Director**

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Non-Technical Summary

The sustainability of US cotton production is pivotal to agricultural sustainability in the Southern US. Protecting and improving soil health, reducing greenhouse gas (GHG) emissions, and enhancing input use efficiency are central to improving the sustainability of cotton in the face of changing climatic conditions. According to the most recent USDA-ERS report on tillage and conservation cropping in the US (Claassen et al. 2018), 60% of cotton acreage is still under conventional tillage, compared for example, to only 30% or less in soybean. Tillage intensity has increased due to the need to control herbicide-resistant and other problematic weeds (CAST 2012), which is now widespread throughout the US Cotton Belt (Heap 2023). Intensive tillage contributes greatly to greenhouse gas (GHG) emissions, stored carbon loss, and a loss of overall soil health, while conservation tillage is considered a vital component of a sustainable soil health management system (Elder and Lal 2008; USDA-NRCS 2017). Cotton is inherently a low residue crop, which in turn adds limited amounts of carbon during the growing season. Because cotton is commonly grown with low residue crops in rotation (Claassen et al. 2018), this is a major area where rapid gains can be achieved in soil health improvement and climate change mitigation. In this regard, cover crops and living mulches can play a key role (Wick et al. 2017). A recent analysis by the Cotton 2040 initiative revealed that cotton is a vulnerable crop to climate change impacts (WTW 2022). There is a crucial need to make cotton cultivation a carbon-neutral, or better yet a net carbon-positive enterprise, while simultaneously mitigating/adapting to the adverse effects of climate change. The majority of US cotton is already grown in marginal soils with low organic matter content. Potential increases in tillage to manage the anticipated surge in herbicide-resistant weeds associated with climate change (Ramesh et al. 2017) is expected to exacerbate this problem. Further, the use of Artificial Intelligence/Machine learning (AI/ML)-assisted precision application technologies can greatly improve resource input use efficiency, which can in turn reduce GHG emissions (e.g. Cotton Inc. 2018). These precision regenerative practices are also expected to create new market opportunities, through, for example, carbon crediting and sustainable cotton markets, and empower the rural agricultural workforce via employment generation.

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The long-term goal of the proposed research is to apply improved precision management practices to enable enhanced carbon sequestration and reduced greenhouse gas emission, pest control, and nutrient and water management; and to address labor challenges, steward land, create new market opportunities, and ensure a sustainable supply of climate-smart cotton. Achieving this will require a coordinated effort among researchers, Extension specialists, and broader industry stakeholders. The specific objectives of this project are: 1. Develop regenerative cotton production practices and investigate their long-term effects on US Belt-wide cotton production systems. 1.1. Establish soil organic carbon and carbon intensity baselines for ecoregions in the US Cotton Belt; 1.2. Investigate regenerative practices for reducing tillage and improving soil health; 1.3. Investigate the long-term impact of regenerative practices in addressing climate change using simulation models. 2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change mitigation/adaptation in cotton. 3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities. 4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt. 5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities. 6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce.

This integrated project is expected to bring tremendous benefits to the US cotton industry through the development and implementation of sound regenerative practices for reducing tillage and addressing climate-change impacts. By utilizing precision agriculture technologies, there will be an improvement in the timely diagnosis of field issues and implementation of site-specific measures for increasing input use efficiency and reducing GHG emissions. New market opportunities will be generated, improving overall farm profitability. The economic analysis will highlight the economic benefits of the adoption of precision regenerative practices and smart-technologies for cotton production. The health and safety of farm workers will be improved. Automation will help alleviate labor shortage in rural areas, while equipping them for new employment opportunities.

Accomplishments

Major goals of the project

The **long-term goal** of the proposed research is to apply improved precision management practices to enable enhanced carbon sequestration and reduced greenhouse gas emission, pest control, and nutrient and water management; and to address labor challenges, steward land, create new market opportunities, and ensure a sustainable supply of climate-smart cotton.

The specific objectives of this project are:

1. Develop regenerative cotton production practices and investigate their long-term effects on US Belt-wide cotton production systems
 - 1.1. Establish soil organic carbon and carbon intensity baselines for ecoregions in the US Cotton Belt
 - 1.2. Investigate regenerative practices for reducing tillage and improving soil health
 - 1.3. Investigate the long-term impact of regenerative practices in addressing climate change using simulation models
2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change mitigation/adaptation in cotton
3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities
4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt
5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities
6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce

What was accomplished under these goals?

Objective 1. Develop regenerative production practices and determine their long-term effects on US cotton.

During the 2024 reporting period, multiple activities were initiated to advance regenerative production practices and begin establishing baseline data to assess long-term impacts. In New Mexico, cotton producers were identified in collaboration with an NMSU Extension Specialist to initiate on-farm engagement. Baseline soil organic carbon and carbon intensity levels are being established for these producers. In Texas, research plots were established to evaluate the use of living mulch using perennial grasses in cotton systems, aiming to improve soil health and reduce inputs. Additionally, 150 cotton cultivars were grown at Auburn University to increase seed availability. In the 2025 season, these cultivars will be planted at Texas A&M University to collect preliminary performance data. Further evaluations will be conducted during the upcoming growing season. To support root system studies and belowground biomass tracking, a custom rhizobox was constructed at Auburn University. In North Carolina, cotton farmers, including those from underrepresented groups, were recruited, and soil samples

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were collected to determine baseline soil organic carbon content. These efforts contribute to establishing regional benchmarks for soil health and tracking improvements over time through regenerative management.

Objective 2. Develop and/or evaluate precision AI/ML and smart technologies for resource conservation and climate change adaptation in cotton.

During the 2024 reporting period, several key research and field activities were conducted to support the development and evaluation of precision and smart technologies in cotton production systems. In center-pivot irrigated cotton fields, circular buffer strips of native perennial grasses were established and evaluated to assess their potential for improving soil health and enhancing ecosystem services. In Mississippi, experimental sites for the summer cover crops were identified and secured, with targeted land preparation completed ahead of the 2025 growing season. These fields reflect regionally relevant cropping histories, having maintained consistent crop rotations, including cotton, for over two decades, which provides an ideal setting for evaluating long-term agricultural interventions. Field monitoring was conducted biweekly to ensure proper maintenance of installed gas chambers used for measuring greenhouse gas fluxes. During these visits, plant height data were recorded to track cotton growth stages. Before harvest, gas chambers were removed, and biomass samples were collected in all four directions from the flux towers, each covering a 1-meter transect. Total aboveground biomass was measured, and seed cotton was separated and weighed to determine cotton yield.

Objective 3. Evaluate the economic feasibility of precision regenerative production practices and determine new market opportunities.

As this was the first year of the project, primary efforts under this objective focused on establishing the foundational framework necessary for future economic evaluations. Initial coordination meetings were held among economists, agronomists, and data scientists to align research trials and data collection with forthcoming economic analyses. This included identifying which regenerative practices, such as cover cropping, conservation tillage, and precision input management, will be evaluated for their economic impacts over the next few years. Data-sharing frameworks are being developed to assess carbon sequestration potential and better understand farmers' experiences and perceptions of emerging market-based incentives. While no comprehensive analysis has been completed yet, these early-stage activities are essential for evaluating cost structures, yield variability, and market opportunities associated with regenerative cotton production in future reporting years.

Objective 4. Expand knowledge about farmers' multi-dimensional experiences of adoption of regenerative practices across the Cotton Belt

To advance understanding of cotton farmers' perspectives on regenerative practices, several outreach and engagement activities were carried out during the 2024 reporting period. Participant observation was conducted at key events to build relationships with growers and better understand their experiences and challenges related to regenerative agriculture. These included two field days and the National Black Growers Council (NBGC) Annual Meeting. The first field day, part of NBGC's 2024 Model Farm Series, was held on August 29-30 at James Farms in Montezuma, Georgia, and focused on cotton, peanuts, and soybeans. The second event, the Cotton Production Field Day organized by the University of Georgia, took place on September 24 at the J. Phil Campbell Research and Education Center in Watkinsville, Georgia. The NBGC Annual Meeting, held in Charleston, South Carolina (November 11-13), allowed the researcher to deepen previously established relationships and build new ones with NBGC members and partner organizations. Additionally, a collaboration was initiated with the Soil Health Institute (SHI) to organize in-person focus groups and collect soil samples in Georgia. SHI developed a targeted list of potential participants based on farming practices, and recruitment efforts are currently underway. These initiatives are key to capturing the diverse and place-based experiences of farmers across the Cotton Belt in adopting regenerative practices.

Objective 5. Promote the adoption of regenerative, climate-smart cotton production practices through innovative and collaborative extension and outreach activities.

In this initial year, project partners engaged in multiple outreach efforts aimed at building awareness and fostering interest in regenerative, climate-smart cotton production practices. Project team members presented research concepts, objectives, and preliminary progress at numerous extension events, including those targeting county extension agents, private crop consultants, natural resource conservation professionals, and academic audiences. These interactions helped disseminate early insights and gather stakeholder feedback to better tailor outreach efforts moving forward. A notable outreach success was achieved through the Mississippi Master Irrigator Program, where formal training sessions were delivered to cotton growers and water managers. These educational engagements are strengthening local capacity and setting the stage for larger-scale adoption efforts planned in future years, including demonstration field days and participatory on-farm trials.

Objective 6. Provide educational opportunities to train the next generation of research and extension scientists and practitioners, including underserved communities, and empower the rural workforce

In 2024, the project began laying the foundation for its educational goals through early curriculum integration and graduate-level instruction. Project investigators began incorporating content on regenerative practices, such as the role of cover crops, root system architecture, and physiological traits related to nutrient and water stress, into university coursework. These efforts aim to provide students with a systems-level understanding of sustainable cotton agroecosystems and prepare them for

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research or extension roles in climate-smart agriculture.

In addition, planning has begun for more comprehensive educational initiatives to be launched in subsequent project years, including summer undergraduate research internships, field-based experiential learning, and the development of online teaching modules. These educational activities will not only build a skilled future workforce, but also foster long-term engagement with rural communities across the Cotton Belt.

What opportunities for training and professional development has the project provided?

During the 2024-2025 reporting period, the project team actively engaged in a wide range of training, outreach, and professional development activities aimed at promoting climate-smart cotton practices and building capacity among stakeholders. These efforts reached a diverse audience, including farmers, ranchers, agricultural consultants, researchers, and industry professionals across multiple states. Project investigators participated in numerous conferences, grower meetings, field days, workshops, and podcasts, providing platforms for disseminating research-based knowledge and practical recommendations.

In New Mexico, Dr. Rajan Ghimire presented at the Innovative Farming Conference in February 2025, engaging 75 farmers, ranchers, and local stakeholders. In Texas, Dr. Ben McKnight led or contributed to multiple major outreach events. These included the BIG Conference in Waco (100 participants), the 2025 Field Crops Symposium in Corpus Christi (65 participants), the Bell County Crops Conference in Temple (115 participants), the Upper Gulf Coast Feed Grain and Cotton Conference in El Campo (150 participants), and the Concho Valley Cotton Conference in San Angelo (200 participants). Additional sessions included a smaller meeting in De Leon with 15 producers and the Southwest Cotton Physiology Meeting in Lubbock, where 150 stakeholders participated. Dr. McKnight also played a significant role in the Texas Plant Protection Association Conference in Bryan, TX, which reached approximately 300 agricultural professionals.

In Mississippi and surrounding states, Dr. Drew Gholson was instrumental in coordinating and presenting at numerous grower meetings, educational courses, and conferences. In February 2025 alone, he delivered presentations at county-level grower meetings in Leflore, Humphreys, Attala/Holmes, Grenada, and Tunica Counties, with each event drawing between 30 to 45 participants. He also led the Mississippi Master Irrigator Course (70 participants), and represented the program at the MS Master Irrigator Multi-State Conference held in Tennessee, which included participants from Arkansas, Louisiana, Mississippi, and Missouri (300+ participants). In addition, Dr. Gholson contributed to two sessions at the Oklahoma State University Winter Crop School in December 2024, addressing irrigation strategies and tillage impacts on soil moisture (each attended by 250 participants). He also participated in the Precision Cotton Research Meeting in Louisiana (80 participants), the 2025 Beltwide Cotton Conferences (40 participants), and the 2024 Mississippi State University Row Crop Short Course, which drew an audience of over 500.

Further professional engagement included presentations at the 51st Annual Mississippi Agricultural Consultants Association Conference (approx. 200 consultants), the Mississippi Agricultural Consultants Association's fall irrigation update meeting (approx. 150 consultants), and the MAIC Row Crop Certified Crop Advisors Program (approx. 300 participants). Dr. Gholson also presented RISER program updates during the MS Cotton Inc. State Support Committee meeting in July 2024. His outreach efforts extended to digital platforms through the Mississippi Crop Situation Podcast in June 2024, where he discussed irrigation triggers for corn, soybean, and cotton, reaching an estimated 100+ listeners.

In addition to these outreach efforts, the project contributed significantly to teaching and student engagement. Across multiple institutions, faculty members mentored both graduate and undergraduate students and incorporated project-related content into classroom teaching. In Georgia, Dr. Nicholas Basinger mentored 21 graduate and 56 undergraduate students while teaching courses such as Weed Science, Weed Science Lab, Integrated Pest Management, and preparing students for experiential weed science competitions. In Alabama, Dr. Alvaro Sanz-Saez supervised one MS student who initiated field sampling and constructed rhizoboxes for project-related research. He also mentored two undergraduate students who supported Fall 2024 experiments by learning field sampling, organization, and image analysis using RhizoVision Explorer. Dr. Sanz-Saez integrated climate-smart cotton research into two graduate-level courses: Principles of Plant Nutrition (CSES7540), focusing on root architecture for nutrient uptake, and Crop Physiology (CSES7250), highlighting the role of root trait variation in improving cultivar water and nutrient use efficiency. Additionally, Dr. Ramon Leon in North Carolina mentored 14 graduate students while teaching courses such as Weed Biology (CS716) and Herbicide Behavior in Plants (CS729). Together, these activities provided substantial opportunities for professional growth, technical training, and stakeholder engagement, reinforcing the project's objectives to advance climate-smart cotton systems through education, research dissemination, and collaborative learning.

How have the results been disseminated to communities of interest?

The research team has actively engaged in multiple outreach and dissemination efforts to share early project findings with stakeholders and communities of interest. In August 2024, team members participated in the Organic Cotton Field Day held in Lamesa, Texas, where initial project objectives, field activities, and the role of regenerative practices in climate-smart cotton production were shared with growers, consultants, extension agents, and other agricultural professionals. In addition to field-based outreach, the team is preparing to present preliminary research results at several national and regional professional

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conferences. These include the American Society of Agronomy Annual Meeting, the Beltwide Cotton Conferences, and the Weed Science Society of America Annual Meeting. Presentations will focus on key aspects of the project, such as soil health assessments, regenerative production trials, precision irrigation approaches, and stakeholder engagement strategies. Furthermore, research and extension personnel have delivered presentations at state-level extension meetings and training sessions, including through programs like the Mississippi Master Irrigator. These engagements are helping to raise awareness of climate-smart cotton practices, encourage farmer participation, and build capacity among agricultural advisors and support personnel. Ongoing efforts are also underway to develop outreach materials, including technical bulletins, training modules, and digital content, which will be disseminated through university extension platforms and partner networks. As the project progresses, a combination of peer-reviewed publications, stakeholder workshops, and on-farm demonstration events will be used to further broaden the reach and impact of project findings.

What do you plan to do during the next reporting period to accomplish the goals?

In the next reporting period, the project team will intensify research, outreach, and education activities to advance progress across all six objectives. Soil sampling efforts will expand across all participating states to establish and strengthen baseline data on soil organic carbon, greenhouse gas emissions, and other soil health indicators. These efforts will include both on-farm and on-station locations representing diverse ecoregions within the Cotton Belt. Controlled experiments on regenerative practices, including cover crops, conservation tillage, and living mulches, will proceed into their second growing season, allowing researchers to gather multi-season agronomic, environmental, and economic data. Field data collection related to crop growth, root traits, ET-based irrigation scheduling, and pest/nutrient diagnostics will support the development of precision AI/ML tools and models for smart cotton management. Gas flux monitoring and remote sensing will also continue to generate valuable information for simulation modeling and resource-use optimization. In parallel, economic analyses will begin using early field data to assess production costs, yield impacts, and the potential value of carbon markets for participating growers. On the outreach front, field days and extension events will be expanded to include on-farm demonstrations. Training programs such as the Mississippi Master Irrigator and educational presentations at stakeholder meetings will continue. The education team will also begin implementing undergraduate summer research programs, developing online learning modules, and integrating cotton sustainability concepts into undergraduate and graduate curricula. Sociological work will advance with the launch of in-person focus groups in Georgia, designed to capture farmers' multi-dimensional experiences with regenerative practice adoption. These efforts will provide critical insights into barriers and opportunities for climate-smart agriculture across the Cotton Belt. Collectively, these planned activities will deepen scientific understanding, strengthen producer engagement, and contribute to achieving the project's long-term sustainability goals.

Participants**Actual FTE's for this Reporting Period**

Role	Non-Students or faculty	Students with Staffing Roles			Computed Total by Role
		Undergraduate	Graduate	Post-Doctorate	
Scientist	4.8	0	0	0	4.8
Professional	0.2	11	4	0.5	15.7
Technical	0.8	0	0	0	0.8
Administrative	0.2	0	0	0	0.2
Other	0	0	0	0	0
Computed Total	6.0	11	4	0.5	21.5

Student Count by Classification of Instructional Programs (CIP) Code

Undergraduate	Graduate	Post-Doctorate	CIP Code
10	3	2	01.00 Agriculture, General.
7	2	1	01.11 Plant Sciences.
3	1	0	01.12 Soil Sciences.

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Undergraduate	Graduate	Post-Doctorate	CIP Code
2	2	0	01.02 Agricultural Mechanization.

Target Audience

The key target audience for this project include cotton farmers throughout the US Cotton Belt, National Cotton Council, Cotton Incorporated, Textile Exchange, US Cotton Trust Protocol, Climate Beneficial Fiber, various cotton merchandisers, Soil Health Institute, University Extension, county extension agents, graduate and undergraduate students, as well as the general public interested in learning sustainable cotton production.

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Yes, this project/program did not produce any scholarly publications during the reporting period and all data assets produced are exempt from DR 1020-006.

Type

Conference Papers and Presentations

Status

Published

Year Published

2025

NIFA Support Acknowledged

YES

Digital Object Identifier (DOI)**Author ORCID(s)****Citation**

Elek Nagy, Gayan Abeysinghe, Sanjay Antony-Babu. Hunting the Hyphosphere: A method for baiting hyphal-dwelling bacteria from soil. At the 36th Annual Texas Plant Protection Conference, College Station TX

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YES

Digital Object Identifier (DOI)**Author ORCID(s)****Citation**

Reyes A, Singh J, Jhonson A, Gamble A, Feng Y, Balckom K, Sanz-Saez A (2025) Impact of Cover Crops on Cotton: Evaluating Root Characteristics and Photosynthetic Traits in Southern and Northern Alabama. Belt Wide Cotton Conference, New Orleans, LA, January 14-16, 2025.

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Type

Conference Papers and Presentations

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2024

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Digital Object Identifier (DOI)**Author ORCID(s)****Citation**

Singh J, Gamble A, Brown S, Koebernick Bartley P, Sanz-Saez A (2024) Exploring Cover Crops in Alabama: Assessing Soil Health, Yield, and Root Characteristics of Following Cash Crop. Southern ASA Meeting, February 3-5, 2024, Atlanta, Georgia.

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Other Products

{Nothing to report}

Changes/Problems

The 2024-25 reporting period marked the first year of coordinated field activities across all participating institutions, and as with the initiation of any large-scale, multi-site research effort, we encountered a number of shared operational and logistical challenges that required adjustments in our approach. A key issue across several sites was the delay in personnel hiring. Although most teams planned to onboard graduate students in Summer 2024, various administrative and hiring delays meant that several students could not begin until the Fall semester. This affected the timing of field experiments and data collection, particularly for weed competition and trait evaluation studies.

Furthermore, it took time to develop detailed protocols, as they involved various nuances. For example, soil sampling is conducted to determine several metrics, some of which required specific sampling methods. Thus, combining all these elements in an efficient manner took time to solidify. This process delayed the onset of the soil sampling campaign across the partner sites. To address this, soil sampling activities will be conducted across two field seasons. Seed procurement for seed increase trials was another cross-cutting issue; due to supply chain and coordination delays, planting occurred later than ideal at several locations. While most sites still achieved good plant growth and viable seed production, the delay added pressure to field timelines.

Despite these initial hurdles, the project has made significant progress. Graduate students are now actively engaged, and foundational work, such as perennial grass screenings, herbicide tolerance assessments, and weed suppression studies, has begun. Importantly, all participating sites have successfully identified and secured field locations, with land preparation completed in readiness for the 2025 growing season. These early efforts and lessons learned will help streamline coordination, improve communication, and strengthen implementation in the coming project years.